

# A Rigid Modular Sieve for Non-Trivial Cycles in the Collatz Dynamics

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## Abstract

*We present a novel framework for excluding non-trivial cycles in the Collatz  $(3n+1)$  problem by coupling global algebraic constraints with local finite field structures. By decomposing the loop accumulator  $R$  into a structural minimum  $R_{\min}(O)$  and an orbit-dependent discrepancy  $\Delta(O)$ , we project the cyclic trajectory onto a rigid congruence lattice over  $F_p$ . This yields a sieve that systematically eliminates parameter pairs  $(O, S)$  based on modular incompatibility, independent of numerical bounds. The method bridges the traditional gap between real-valued logarithmic estimates and 2-adic local analyses, offering a new arithmetic invariant for cycle exclusion.*

## 1. Introduction

The Collatz map  $T: \mathbb{N} \rightarrow \mathbb{N}$  is defined for odd integers as  $T(n) = (3n+1)/2^{v_2(3n+1)}$ . Traditional exclusion methods fall into two disconnected categories: Real-domain bounds (size bounds, numerical dependency) and 2-adic local constraints (analyze on  $\mathbb{Z}_2$  but suffer from Haar singularity). This paper introduces P34, a rigid modular sieve that bridges this gap by projecting the global cycle identity onto a finite field  $F_p$  while retaining the full algebraic structure of the orbit.

## 2. The Classical Cycle Identity

For a cycle of odd length  $O$  and total step length  $S$ , the starting point  $n_0$  satisfies the exact algebraic identity:

$$2^S n_0 = 3^O n_0 + R$$

where  $R = \sum_{j=0}^{O-1} 3^j 2^{e_j}$ . From this, the divisibility constraint follows:  $n_0 \mid (2^S - 3^O)$ , and equivalently  $n_0(2^S - 3^O) = R$ .

## 3. Decomposition of the Accumulator

We introduce a canonical decomposition of  $R$ :

$$R_{\min}(O) = (6^O - 1) / 5$$

which is the algebraic lower bound obtained by setting  $e_j = j$ .

Define the discrepancy:  $\Delta(O) = R - R_{\min}(O) = \sum_{j=0}^{O-1} 3^j (2^{e_j} - 2^j)$ .

The cyclic identity becomes:  $n_0(2^S - 3^O) = R_{\min}(O) + \Delta(O)$ .

## 4. The Modular Sieve Projector

Projecting onto the finite field  $F_p$  for an arbitrary prime  $p$ , any valid cycle configuration must satisfy:

$$n_0 \cdot (2^S - 3^O) \equiv R_{\min}(O) + \Delta(O) \pmod{p}$$

with the global boundary constraint:  $n_0 \leq (2^S - 3^O) / (R_{\min}(O) + \Delta(O))$ .

## 5. The Cycle-Exclusion Criterion (P34)

**Theorem (P34, Cyclic Congruence Sieve):** If a non-trivial cycle exists, then for every prime  $p$ , there exist integers  $n_0, O, S$  satisfying:

1. The congruence projection:  $n_0(2^S - 3^O) \equiv R_{\min}(O) + \Delta(O) \pmod{p}$
2. The inequality constraint:  $n_0 \leq (2^S - 3^O) / (R_{\min}(O) + \Delta(O))$
3. The lower bound  $S/O > \log_3 3$  and upper bound  $S/O < \log_3 3 + 1/O$ .

## 6. Sieve Efficacy and Computational Implications

The sieve can be applied without numerical iteration over  $n_0$ . For each  $O$ , compute  $R_{\min}(O) \pmod{p}$ . For each candidate  $S$  within the logarithmic bounds, test whether there exists  $n_0$  satisfying both congruence and inequality. If no  $n_0$  exists, eliminate that  $(O, S)$  pair.

## 7. Connection to the Discrepancy Distribution (M7)

The sieve's strength depends on the distribution of  $\Delta(O) \pmod{p}$ . Under the natural density assumption (M7),  $\Delta(O) \pmod{p}$  becomes uniformly distributed as  $O \rightarrow \infty$ . This implies that the fraction of  $(O, S)$  pairs passing the sieve decays as  $O(p^{-m})$  when multiple primes are used.

## 8. Conclusion and Outlook

P34 provides the first algebraic sieve that uses exact cycle identities, operates in finite fields, and couples global cycle parameters with local arithmetic invariants.

## References

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